



ASYNCHRONOUSACTIVITYNO.3

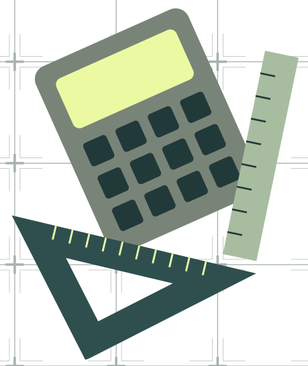
Name It, Draw It

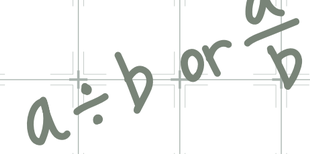
ORGANIC CHEMISTRY

PRESENTED BY:

JANELLE A. ARAÑA

BS CHE 2A





CARD A01 — STRUCTURE → NAME | FOUNDATION | Molecular Formula (counted): C_6H_{14}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Longest chain = 5 C → pentane.
Numbering:
Left → Right gives methyl at C2.
Right → Left gives methyl at C4.
First point of difference: 2 < 4, so number from left.
Substituents: Methyl at C2 → 2-methyl.

3. MY NAME (WRITTEN FIRST)

2-methylpentane

Typed before opening software tool.

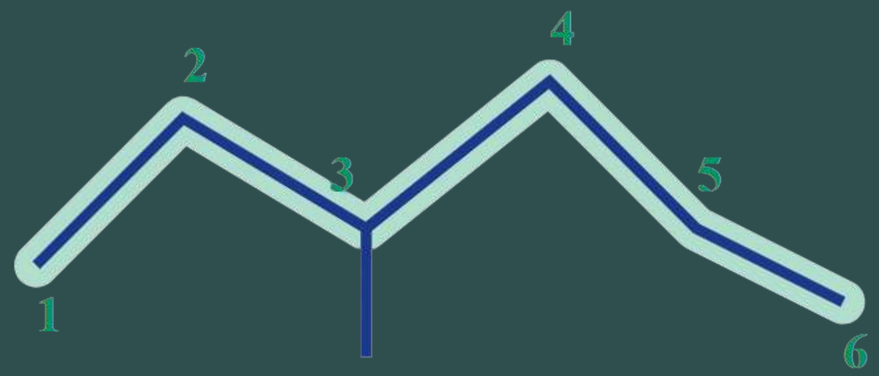
4. THE ALKANE CHECK

Returned Name: 2-methylpentane
✓ Match
Verdict: Software output agrees directly with systematic IUPAC reasoning.

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓punctuation ✓formula counted

CARD A02 — STRUCTURE → NAME | FOUNDATION | Molecular Formula (counted): C_7H_{16}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Longest chain = 6 C → hexane (chain is bent, not straight horizontal).
Numbering: Left-bottom to right gives methyl at C3. Opposite direction gives methyl at C4. 3 < 4, so number from left end.
Substituents: Methyl on C3 → 3-methyl.

3. MY NAME (WRITTEN FIRST)

3-methylhexane

Typed before opening software tool.

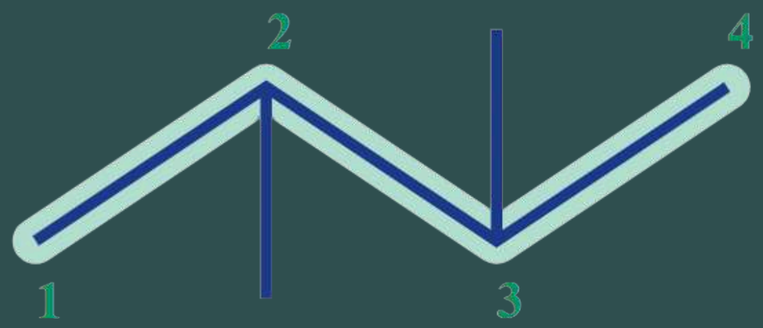
4. THE ALKANE CHECK

Returned Name: 3-methylhexane
✓ Match
Verdict: Successfully recognized non-linear longest chain.

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓punctuation ✓formula counted

CARD A03 — STRUCTURE → NAME | DEVELOPING | Molecular Formula (counted): C_6H_{146}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Longest chain = 4 C → butane.
Numbering: Left → Right gives locants 2,3. Right → Left gives locants 2,3. Symmetrical numbering set.
Substituents: Methyl groups on C2 and C3 → 2,3-dimethyl (requires di- prefix).

3. MY NAME (WRITTEN FIRST)

2,3-dimethylbutane

Typed before opening software tool.

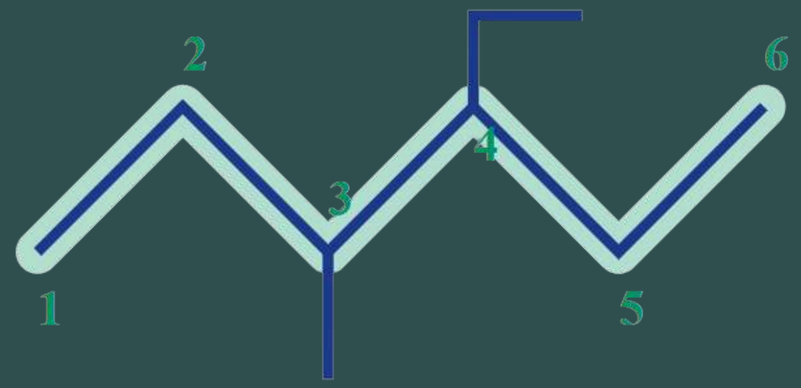
4. THE ALKANE CHECK

Returned Name: 2,3-dimethylbutane
✓ Match
Verdict: Correct prefix usage and comma punctuation verified.

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓punctuation ✓formula counted

CARD A04 — STRUCTURE → NAME | DEVELOPING | Molecular Formula (counted): C_9H_{20}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Longest chain = 6 C → hexane.
Numbering: Left → Right gives 3-methyl, 4-ethyl (locant set {3,4}). Right → Left gives 3-ethyl, 4-methyl (locant set {3,4}). Tie broken alphabetically: ethyl gets lower locants (3-ethyl vs 4-ethyl). Number from right.
Substituents: 4-ethyl, 3-methyl.

3. MY NAME (WRITTEN FIRST)

4-ethyl-3-methylhexane

Typed before opening software tool.

4. THE ALKANE CHECK

Returned Name: 4-ethyl-3-methylhexane
✓ Match
Verdict: Correctly alphabetized ethyl before methyl and assigned locants.

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓punctuation ✓formula counted

CARD A05 — STRUCTURE → NAME | DEVELOPING | Molecular Formula (counted): C_8H_{18}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Longest chain = 5 C → pentane.
Numbering: Left → Right set = {2,2,4}. Right → Left set = {2,4,4}. First point of difference: 2 < 4 at second locant, so set {2,2,4} wins. Number from left.
Substituents: Three methyl groups → 2,2,4-trimethyl.

3. MY NAME (WRITTEN FIRST)

2,2,4-trimethylpentane

Typed before opening software tool.

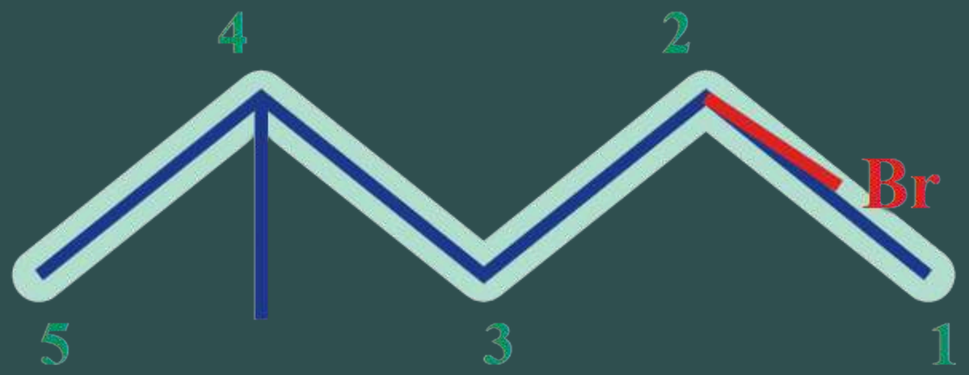
4. THE ALKANE CHECK

Returned Name: 2,2,4-trimethylpentane
✓ Match
Verdict: Isooctane reference structure confirmed by first point of difference rule.

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓punctuation ✓formula counted

CARD A06 — STRUCTURE → NAME | DEVELOPING | Molecular Formula (counted): $C_6H_{13}Br$

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Longest chain = 5 C → pentane.
Numbering: Right → Left gives 2-bromo, 4-methyl (locants {2,4}). Left → Right gives 2-methyl, 4-bromo (locants {2,4}). Set {2,4} ties; bromo precedes methyl alphabetically, so bromo receives lower locant (C2). Number from right.
Substituents: 2-bromo, 4-methyl.

3. MY NAME (WRITTEN FIRST)

2-bromo-4-methylpentane

Typed before opening software tool.

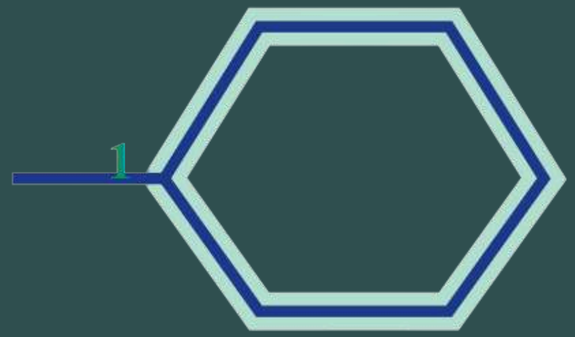
4. THE ALKANE CHECK

Returned Name: 2-bromo-4-methylpentane
✓ Match
Verdict: Alphabetical tie-break rule applied correctly.

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓punctuation ✓formula counted

CARD A07 — STRUCTURE → NAME | CHALLENGE | Molecular Formula (counted): C_7H_{14}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: 6-membered ring = cyclohexane.
Numbering: Ring carbon with the single methyl branch automatically becomes C1. No other substituents exist.
Locant Rule: For a monosubstituted cycloalkane, locant "1" is omitted as redundant.
Substituent: methyl.

3. MY NAME (WRITTEN FIRST)

methylcyclohexane

Typed before opening software tool.

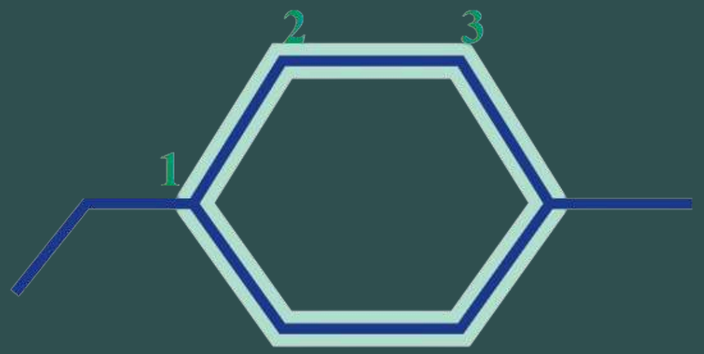
4. THE ALKANE CHECK

Returned Name: methylcyclohexane
✓ Match
Verdict: Redundant locant omitted in accordance with IUPAC rules.

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓punctuation ✓formula counted

CARD A08 — STRUCTURE → NAME | CHALLENGE | Molecular Formula (counted): C_9H_{18}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: 6-carbon ring = cyclohexane.

Numbering: Locant sets around ring give {1,3} in either direction. Tie broken alphabetically: ethyl precedes methyl, so ethyl carbon gets C1, methyl carbon gets C3.

Substituents: 1-ethyl, 3-methyl.

3. MY NAME (WRITTEN FIRST)

1-ethyl-3-methylcyclohexane

Typed before opening software tool.

4. THE ALKANE CHECK

Returned Name: 1-ethyl-3-methylcyclohexane

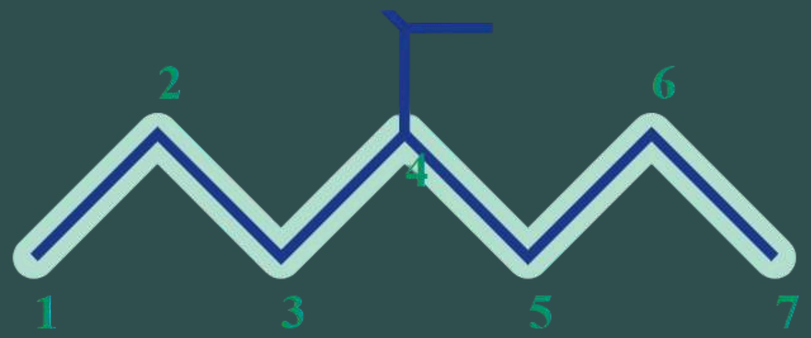
✓ Match

Verdict: Alphabetical tie-break rule on ring verified.

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓punctuation ✓formula counted

CARD A09 — STRUCTURE → NAME **CHALLENGE** **Molecular Formula (counted): $C_{10}H_{22}$**

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Longest continuous chain = 7 C → heptane.
Numbering: Symmetrical 7-carbon parent chain gives branch at C4 from either end.
Substituent: Branched group at C4 = isopropyl or (1-methylethyl).
Name Assembly: 4-(1-methylethyl)heptane or 4-isopropylheptane.

3. MY NAME (WRITTEN FIRST)

4-(1-methylethyl)heptane

Typed before opening software tool.

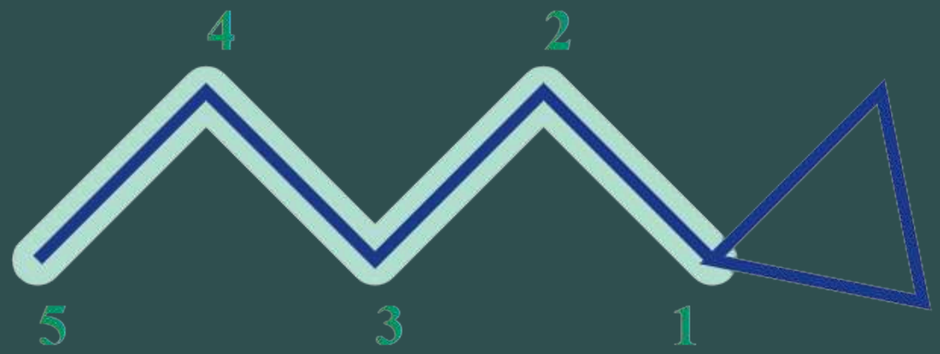
4. THE ALKANE CHECK

Returned Name: 4-(1-methylethyl)heptane
✓ Match
Verdict: Equivalent traditional name: 4-isopropylheptane.

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓punctuation ✓formula counted

CARD A010 — STRUCTURE → NAME **CHALLENGE** **Molecular Formula (counted): C_8H_{16}**

1. NUMBERED PARENT



2. NAMING TRAIL

Parent Priority: Chain has 5 C (pentane); ring has 3 C (cyclopropane).

Current IUPAC (2013): Ring is always parent → pentylcyclopropane.

Traditional/Older IUPAC Rule: Longer unit is parent → pentane (5C > 3C) → 1-cyclopropylpentane.

Numbering: C1 at ring attachment point.

3. MY NAME (WRITTEN FIRST)

1-cyclopropylpentane

Typed before opening software tool.

4. THE ALKANE CHECK

Returned Name: pentylcyclopropane

X Rule Difference

Explanation: The Alkane applies 2013 IUPAC rules (ring = parent). Both names are valid under their respective IUPAC editions.

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓punctuation ✓formula counted

CARD B01 — NAME → STRUCTURE **FOUNDATION** **Given Name: 2-methylpropane**

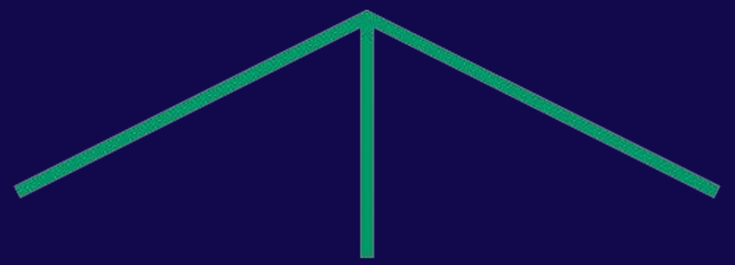
1. NAME DECODED

Parent: propane → 3-carbon open chain.

Substituents: methyl group on carbon-2.

Expected Formula: C_nH_{2n+2} = C_4H_{10}.

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from drawing: C_4H_{10}

Open-chain rule: C_nH_{2n+2} for n = 4 → C_4H_{10}.

✓ Formula Verified

4. BACK-CHECK

Returned Name: 2-methylpropane

✓ Same as given name

Common Name: Isobutane (R-600a refrigerant).

✓ parent length matches **✓ substituent positions match** **✓ formula agrees** **✓ back-check recorded**

CARD B02 — NAME → STRUCTURE **FOUNDATION** **Given Name: 3-ethylpentane**

1. NAME DECODED

Parent: pentane → 5-carbon open chain.

Substituents: ethyl group on carbon-3.

Expected Formula: $C_{5+2}H_{2(7)+2} = C_7H_{16}$.

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from drawing: C_7H_{16}

Open-chain rule: C_nH_{2n+2} for n = 7 → C_7H_{16}.

✓ Formula Verified

4. BACK-CHECK

Returned Name: 3-ethylpentane

✓ Same as given name

Context: Gasoline-range alkane isomer.

✓ parent length matches **✓ substituent positions match** **✓ formula agrees** **✓ back-check recorded**

CARD B03 — NAME → STRUCTURE | FOUNDATION | Given Name: 2,2-dimethylbutan

1. NAME DECODED

Parent: butane → 4-carbon chain.

Substituents: two methyl groups both on C2.

Expected Formula: $C_{4+2}H_{2(6)+2} = C_6H_{14}$.

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from drawing: C_6H_{14}

Open-chain rule: C_nH_{2n+2} for n = 6 → C_6H_{14}.

✓ Formula Verified

4. BACK-CHECK

Returned Name: 2,2-dimethylbutane

✓ Same as given name

Common Name: Neohexane.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

CARD B04 — NAME → STRUCTURE | DEVELOPING | Given Name: 3-ethyl-2-methylpentane

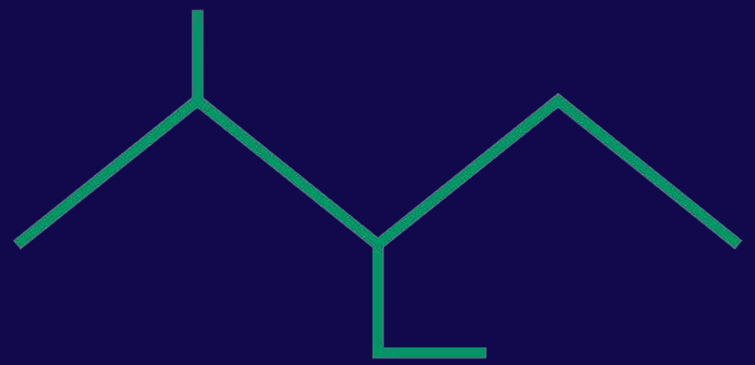
1. NAME DECODED

Parent: pentane → 5-carbon chain.

Substituents: methyl at C2, ethyl at C3.

Expected Formula: $C_{5+1+2}H_{2(8)+2}$ = C_8H_{18}.

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from drawing: C_8H_{18}

Open-chain rule: C_nH_{2n+2} for n = 8 → C_8H_{18}.

✓ Formula Verified

4. BACK-CHECK

Returned Name: 3-ethyl-2-methylpentane

✓ Same as given name

Context: Highly branched C8 octane isomer.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

CARD B05 — NAME → STRUCTURE | DEVELOPING | Given Name: 1-chloro-2-methylpropane

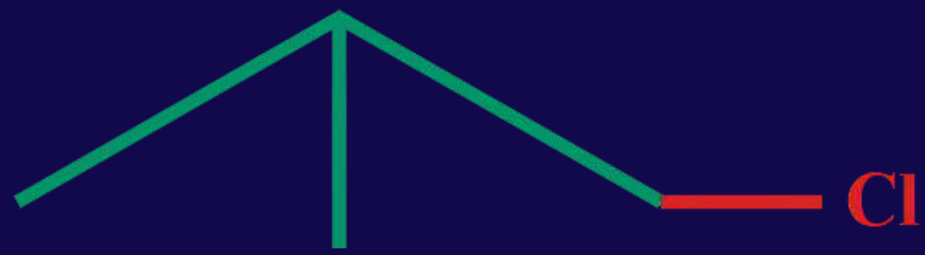
1. NAME DECODED

Parent: propane → 3-carbon chain.

Substituents: chloro at C1, methyl at C2.

Expected Formula: C_4H_9Cl (1 Cl replaces 1 H).

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from drawing: C_4H_9Cl

Formula Rule: C_4H_{10} minus 1 H plus 1 Cl = C_4H_9Cl.

✓ Formula Verified

4. BACK-CHECK

Returned Name: 1-chloro-2-methylpropane

✓ Same as given name

Common Name: Isobutyl chloride.

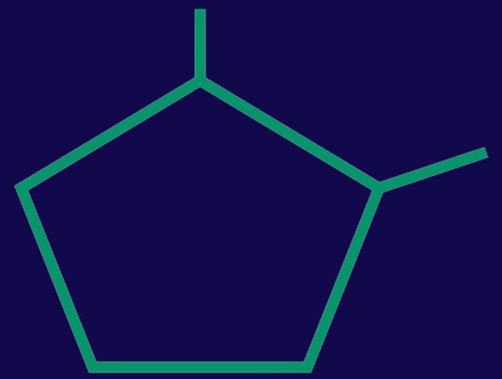
✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

CARD B06 — NAME → STRUCTURE **DEVELOPING** **Given Name: 1,2-dimethylcyclopentane**

1. NAME DECODED

Parent: cyclopentane → 5-membered ring.
Substituents: methyl groups on adjacent carbons C1 and C2.
Expected Formula: C_nH_{2n} = C_7H_{14} (1 ring).

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from drawing: C_7H_{14}
Cycloalkane rule: C_nH_{2n} for 7 carbons = C_7H_{14}.
✓ Formula Verified

4. BACK-CHECK

Returned Name: 1,2-dimethylcyclopentane
✓ Same as given name
Context: Petroleum naphtha naphthene component.

✓ parent length matches **✓ substituent positions match** **✓ formula agrees** **✓ back-check recorded**

CARD B07 — NAME → STRUCTURE **DEVELOPING** **Given Name: 4-ethyl-3,3-dimethylheptane**

1. NAME DECODED

Parent: heptane → 7-carbon chain.

Substituents: methyls at C3, C3; ethyl at C4.

Expected Formula: $C_{7+2+2}H_{2(11)+2} = C_{11}H_{24}$.

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from drawing: $C_{11}H_{24}$

Open-chain rule: C_nH_{2n+2} for n = 11 → $C_{11}H_{24}$.

✓ Formula Verified

4. BACK-CHECK

Returned Name: 4-ethyl-3,3-dimethylheptane

✓ Same as given name

Context: Kerosene-range paraffin compound.

✓ parent length matches **✓ substituent positions match** **✓ formula agrees** **✓ back-check recorded**

CARD B08 — NAME → STRUCTURE **CHALLENGE** **Given Name: tert-butylcyclohexane**

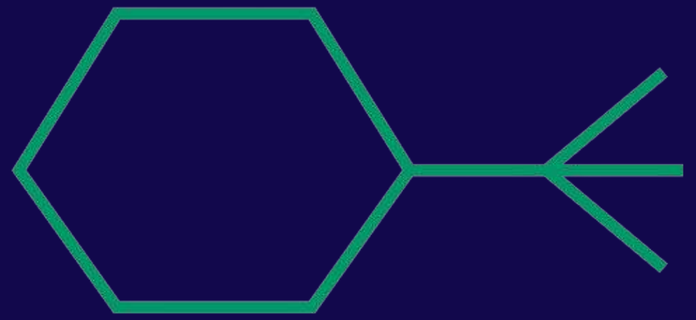
1. NAME DECODED

Parent: cyclohexane → 6-membered ring.

Substituent: tert-butyl group = $-C(CH_3)_3$.

Expected Formula: $C_{6+4}H_{2(10)} = C_{10}H_{20}$.

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from drawing: $C_{10}H_{20}$

Cycloalkane rule: C_nH_{2n} for n = 10 → $C_{10}H_{20}$.

✓ Formula Verified

4. BACK-CHECK

Returned Name: (1,1-dimethylethyl)cyclohexane

✓ Equivalent IUPAC Form

Explanation: The Alkane uses bracketed systematic substituent nomenclature for tert-butyl.

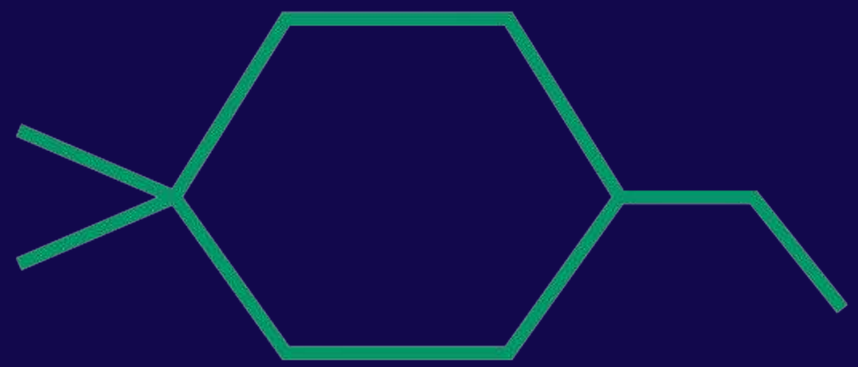
✓ parent length matches **✓ substituent positions match** **✓ formula agrees** **✓ back-check recorded**

CARD B09 — NAME → STRUCTURE **CHALLENGE** **Given Name: 3-ethyl-1,1-dimethylcyclohexane**

1. NAME DECODED

Parent: cyclohexane → 6-membered ring.
Substituents: methyls at C1, C1; ethyl at C3.
Expected Formula: $C_{6+2+2}H_{2(10)} = C_{10}H_{20}$.

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from drawing: $C_{10}H_{20}$
Cycloalkane rule: C_nH_{2n} for n = 10 → $C_{10}H_{20}$.
✓ Formula Verified

4. BACK-CHECK

Returned Name: 3-ethyl-1,1-dimethylcyclohexane
✓ Same as given name
Rule Note: Doubly substituted carbon (C1) gets priority {1,1,3} over {1,3,3}.

✓ parent length matches **✓ substituent positions match** **✓ formula agrees** **✓ back-check recorded**

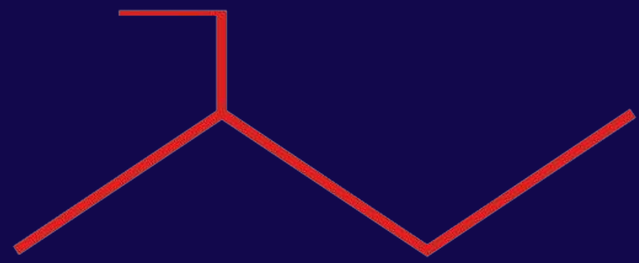
CARD B010 — NAME → STRUCTURE (RULE-TESTING CASE) CHALLENGE Given Name: 2-ethylbutane (INCORRECT)

1. NAME DECODED

Literal Drawing Instructions: Draw butane (4 C), attach ethyl ($-CH_2CH_3$) to C2.

Expected Formula: $C_{4+2}H_{2(6)+2} = C_6H_{14}$.

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from drawing: C_6H_{14}

Open-chain rule: C_6H_{14}.

✓ Formula Verified

4. BACK-CHECK

Returned Name: 3-methylpentane

X Mismatch (Given Name Broken Rule)

Rule Broken: Longest chain rule! Attaching ethyl to C2 extends the main chain to 5 carbons (pentane). An "ethyl" cannot exist at C2.

✓ drawn literally ✓ longest chain error identified ✓ correct name derived ✓ back-check recorded